1 Moments

Discrète		Continue	
Espérance $\mathbb{E}[X]$	**Varérance $\mathbb{V}(X) \geq 0$**	**Espérance $\mathbb{E}[X]$**	**Varérance $\mathbb{V}(X) \geq 0$**
$\sum_{i=1}^{n} x_i \mathbb{P}_X(x_i)$	$\sum_{i=1}^{n} \left(x_i - \mathbb{E}[X]\right)^2 \mathbb{P}_X(x_i)$	$\int_{-\infty}^{+\infty} t f_X(t) dt$	$\int_{-\infty}^{+\infty} \left(t - \mathbb{E}[X]\right)^2 f_X(t) dt$
Linéarité de l'espérance	**Non-linéarité de la variance**	**Linéarité de l'espérance**	**Non-linéarité de la variance**
$\mathbb{E}[aX+b] = a\mathbb{E}[X] + b$	$\mathbb{V}(aX+b) = a^2\mathbb{V}(X)$	$\mathbb{E}[aX+b] = a\mathbb{E}[X] + b$	$\mathbb{V}(aX+b) = a^2\mathbb{V}(X)$
Inégalité de Markov v.a.d. et v.a.c.		**Inégalité de Tchebyshev v.a.d. et v.a.c.**	
$\forall a > 0, \quad \mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$		$\mathbb{P}\left(\lvert X - \mathbb{E}[X]\rvert \geq a\right) \leq \frac{\mathbb{V}(X)}{a^2}$	
Formule de transfert		**Formule de transfert**	
$\mathbb{E}[g(X)] = \sum_{i=1}^{n} g(x_i)\mathbb{P}_X(x_i)$		$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(t) f_X(t) dt$	

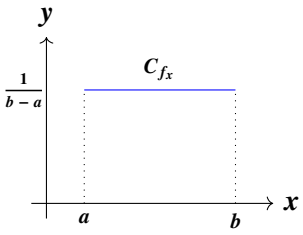
☛ **Comment calculer l'espérance, l'écart-type d'une variable aléatoire continue ?**

On suppose connue f la densité de probabilité de X. Pour l'espérance mathématique : $E(X) = \int_{-\infty}^{+\infty} x f(x) \mathrm{d}x$.

Pour l'écart-type. Calculer : ❶ $E\left(X^2\right) = \int_{-\infty}^{+\infty} x^2 f(x) \mathrm{d}x$, ❷ $V(X) = E\left(X^2\right) - E(X)^2$, ❸ enfin $\sigma(X) = \sqrt{V(X)}$.

2 Lois usuelles

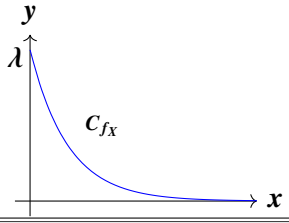
Loi uniforme sur $[a,b]$ $\left(X \hookrightarrow \mathcal{H}_{[a,b]}\right) \implies \left(Z := \frac{X-a}{b-a} \hookrightarrow \mathcal{H}_{[0,1]}\right)$



$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{si } x \in [a,b] \\ 0 & \text{sinon} \end{cases} \quad ; \quad F_X(x) = \begin{cases} 0 & \text{si } x < a \\ \frac{x-a}{b-a} & \text{si } x \in [a,b] \\ 1 & \text{si } x > b \end{cases}$$

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \mathbb{V}(X) = \frac{(b-a)^2}{12} \quad \text{et} \quad \sigma = \frac{b-a}{2\sqrt{3}}$$

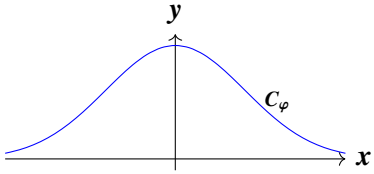
Loi exponentielle $\left(X \hookrightarrow \mathcal{E}(\lambda)\right) \quad (Z := \lambda X \hookrightarrow \mathcal{E}(1))$



$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{si} \quad x \geq 0 \\ 0 & \text{si} \quad x < 0 \end{cases} \quad ; \quad F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{si} \quad x \geq 0 \\ 0 & \text{si} \quad x < 0 \end{cases}$$

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \mathbb{V}(X) = \frac{1}{\lambda^2} \quad \text{et} \quad \sigma = \frac{1}{\lambda}$$

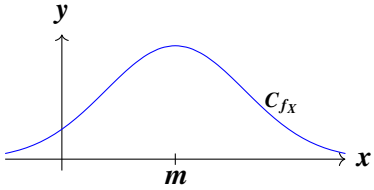
Loi normale centrée réduite $\left(Z \hookrightarrow \mathcal{N}(0,1)\right)$



$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \; ; \quad \Phi(x) = \mathbb{P}(Z \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\mathbb{E}[X] = 0, \quad \mathbb{V}(X) = 1 \quad \text{et} \quad \sigma = 1$$

Loi normale $\left(X \hookrightarrow \mathcal{N}(m,\sigma^2)\right)$ $X \hookrightarrow \mathcal{N}(m;\sigma^2) \implies Z = aX + b \hookrightarrow \mathcal{N}(am+b; a^2\sigma^2)$ et $\left(Z := \frac{X-m}{\sigma} \hookrightarrow \mathcal{N}(0;1)\right)$



$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}} \; ; \quad F_X(x) = \mathbb{P}\left(\frac{X-m}{\sigma} \leq \frac{x-m}{\sigma}\right) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$\mathbb{E}[X] = m \quad \text{et} \quad \mathbb{V}(X) = \sigma^2$$